

Modulation and Power Losses of a Stacked Polyphase Bridge Converter

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Abstract—This paper presents and analyzes the modulation and power losses of a stacked polyphase bridge (SPB) converter—a modular topology potentially allowing increased compactness and fault tolerance for automotive drive systems. In particular, an analysis of the impact of modulation on the harmonic content of the dc link current of the converter is made. Due to the possibility of using higher switching frequency devices, the current ripple of the SPB converter can be smaller than that of the two-level three-phase converter with the same dc link capacitor. However, higher switching frequency will result in higher switching losses per fundamental period. Therefore, this paper also investigates the reduction in the dc link ripple current achieved by interleaving the modulation among the different converter submodules. The amount of reduction can be predicted using an analytical model whose results are demonstrated to be in good agreement with the results from the corresponding simulations and an experimental setup. Furthermore, based on the experimentally verified analysis, the SPB converter in a typical automotive drive system is discussed and compared with a conventional two-level three-phase converter.

Index Terms—DC link ripple current, modular topology, power losses, stacked polyphase bridge (SPB) converter.

I. INTRODUCTION

MODULAR power conversion has received increasing attention in recent years [1], [2]. Trends in power semiconductor technology indicate a tradeoff in the selection of devices in terms of voltage-blocking capability and switching frequency [3], [4]. By lowering the voltage level of each submodule, modular converters can utilize a large variety of semiconductor devices, such as insulated-gate bipolar transistors (IGBTs), metal–oxide–semiconductor field-effect transistors (MOSFETs), and integrated gate-commutated thyristors. The modular multilevel converter (MMC), first presented in [5], is one of the most successfully commercialized modular converter topologies and has been applied in a wide range of high-power applications [6]–[9]. However, the MMC is not particularly suitable for low-power and low-voltage applications. In [10], the modular high-frequency (MHF) converter was proposed for future electric drive systems. The MHF

converter preserves some of the superior features of the MMC but allows for low power and low voltage applications using low-voltage, high-switching-frequency FETs. The stacked polyphase bridge (SPB) converter topology (illustrated in Fig. 1), which combines some of the attractive features of a conventional two-level three-phase converter and the MHF converter, was proposed relatively recently. It has been found suitable for integrated electric drives [11]–[13] and proposed for a transformerless power conversion system for wind turbine applications [14], [15].

In a typical drive system where one or more switched converter are connected to a common dc bus, high-frequency voltage and current harmonics arise due to switching events in the converter. These harmonics will result in dc link current ripple, which gives rise to possible battery life reduction [16], [17]. Basically, there are two ways to reduce this ripple: to increase the switching frequency or increase the size of the passive components. The SPB converter provides the possibility to use a relatively high switching frequency (≥ 100 kHz), avoiding unoptimized component sizing, which could restrain the compactness of a drive system. However, the switching frequency cannot be increased without considering the increased switching losses, which significantly affect the converter efficiency and thermal design. On the other hand, optimized modulation methods can result in less harmonic content, and thus in this way, the switching frequency can be reduced to reach a certain ripple level [18]–[20].

From the discussion above, this paper starting from the theoretical analysis of switching losses and the harmonic content of the dc link current with different modulation techniques introduces an accurate model to estimate the required switching frequency, modulation method, and energy storage capability of the converter. The presented harmonic model and loss model are experimentally verified on a scaled-down prototype. Then, based on the above model, a comparison of the SPB converter and the two-level three-phase converter in a typical automotive application is presented, thoroughly investigating the inherent influence between the ripple level, the power losses, and the energy storage capability of the dc link capacitors.

The outline of this paper is as follows. In Section II, the basic features of the SPB converter are explained. The operating principles, modulation, and harmonics are analyzed in Section III. Section IV provides a description of the switching process and its analytical model. The experimental harmonic spectrum of the dc link current is compared with the theory and simulation results, which are in good agreement,

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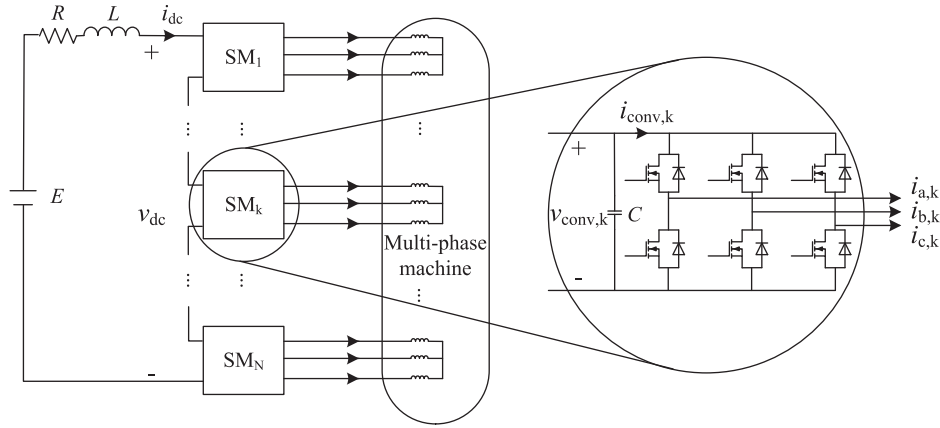


Fig. 1. Structure of the SPB converter topology.

in Section V, where the loss model is also verified by the experimental results. In Section VI, a comparison between the SPB converter and the two-level three-phase converter in an automotive drive system is presented. Finally, conclusions are drawn in Section VII.

II. BASIC FEATURES OF THE SPB CONVERTER

The SPB converter consists of a number of series connected *submodules*, each comprising a conventional three-phase bridge with its associated dc link capacitor. The (three-phase) ac terminal of each submodule can be connected to a set of three-phase windings of a multiphase machine [21]. Since the series connected submodules are sharing the dc voltage, care must be taken so that submodule capacitor-voltage balance is maintained under both normal and fault conditions. Compared with conventional two-level three-phase converters, the SPB converter has the following advantages.

1) *Reduction in Submodule Capacitor Size*: Since the submodules are series connected, the use of low-voltage MOSFETs becomes possible allowing for high switching frequencies, effectively reducing the required submodule capacitor size, which can be further reduced by adopting the interleaving technique presented in Section III.

2) *Scalability*: The modular design of the SPB converter allows for scalability. Its topology can easily be extended to different power and voltage levels, which must be considered an important benefit.

3) *Redundancy*: The SPB converter possesses an inherent fault-tolerant capability. For a converter with N submodules, with proper control after bypassing a single faulty submodule, the input voltage of submodule k , $v_{conv,k}$ increases to $v_{dc}/(N-1)$, where v_{dc} is the total input voltage on the dc side (see Fig. 1). Since semiconductor devices are usually rated with a 50%–100% voltage margin, this voltage rise will generally not destroy the remaining devices. In spite of one submodule failure, the converter can operate without shutting down, improving in this way the system reliability.

4) *Integration*: The use of low-voltage MOSFETs and very small nonelectrolytic capacitors (such as low-voltage film capacitors or possibly ceramic capacitors) enables the integration of the power circuit and control circuit on one printed

circuit board. Moreover, the integration of the converter and the machine allows for short connection leads with reduced cabling costs and potential advantages in terms of weight and electromagnetic interference. These benefits make the SPB converter suitable for EV and HEV drivetrains [11].

III. HARMONIC ANALYSIS OF THE DC LINK CURRENT

An analytical model for predicting the resulting harmonic content in the dc link current of the SPB converter is presented in this section.

By replacing the k th ($k = 1, 2, \dots, N$) submodule converter input current with a current source $i_{conv,k}$, the dynamics of the SPB converter can be expressed as

$$L \frac{di_{dc}}{dt} = E - Ri_{dc} - \sum_{k=1}^N v_{conv,k} \quad (1)$$

$$C \frac{dv_{conv,k}}{dt} = i_{dc} - i_{conv,k} \quad (2)$$

where E is the battery voltage, i_{dc} is the battery current, and L and R are the total battery and cable inductance and resistance, respectively.

Using superposition, from (1) and (2), the expression for i_{dc} at frequency ω is obtained as

$$I_{dc}(\omega) = \frac{\sum_{k=1}^N I_{conv,k}(\omega)}{N - \omega^2 LC + j\omega CR}. \quad (3)$$

Here, $I_{conv,k}(\omega)$ represents the amplitude of $i_{conv,k}$ at frequency ω .

Obviously, from (3), if some of the harmonic components in $\sum_{k=1}^N I_{conv,k}$ can be eliminated, the corresponding harmonics in i_{dc} will also be eliminated.

Considering the submodule structure illustrated in Fig. 1, assuming ideal switches, the time-varying binary switching function $s_{x,k}$ of the upper switch in phase leg x ($x \in \{a, b, c\}$) of submodule k ($k = 1, 2, \dots, N$) can be expressed as

$$s_{x,k} = \begin{cases} 1, & \text{If the upper switch is on; the lower one is off} \\ 0, & \text{If the lower switch is on; the upper one is off.} \end{cases} \quad (4)$$

Furthermore, the ac output currents $i_{a,k}$, $i_{b,k}$, and $i_{c,k}$ are assumed to be sinusoidal and symmetrical ($i_{a,k} + i_{b,k} + i_{c,k} = 0$)

with power factor $\cos \theta_0$. This means that the harmonic ripple components in the ac currents are not considered in this analysis and there is no common mode ac current in the circuit. From (4), the submodule converter input current $i_{\text{conv},k}$ can be expressed as

$$i_{\text{conv},k} = s_{a,k}i_{a,k} + s_{b,k}i_{b,k} + s_{c,k}i_{c,k}. \quad (5)$$

Here, $i_{a,k} = \hat{I} \cos(\omega_0 t + \theta_0)$, $i_{b,k} = \hat{I} \cos(\omega_0 t - (2\pi/3) + \theta_0)$ and $i_{c,k} = \hat{I} \cos(\omega_0 t - (4\pi/3) + \theta_0)$, where $\hat{I}$ and ω_0 are the magnitude and frequency of the (balanced) ac current, respectively.

For carrier-based modulations, the general form of the switching function $s_{a,k}$ is periodic in both carrier wave angular frequency ω_c and the reference frequency ω_0 . Therefore, it can be expanded as a double Fourier series [22]

$$s_{a,k}(t) = \frac{A_{00}}{2} + \sum_{n=1}^{+\infty} [A_{0n} \cos(n[\omega_0 t + \theta_0]) + B_{0n} \sin(n[\omega_0 t + \theta_0])] + \sum_{m=1}^{+\infty} \sum_{n=-\infty}^{+\infty} [A_{mn} \cos(m[\omega_c t + \theta_{c,k}] + n[\omega_0 t + \theta_0]) + B_{mn} \sin(m[\omega_c t + \theta_{c,k}] + n[\omega_0 t + \theta_0])] \quad (6)$$

where m and n represent the carrier and sideband index variables, respectively, $\theta_{c,k}$ is the initial angle of the carrier waveform of submodule k , and A_{mn} and B_{mn} are the real part and imaginary part of the complex Fourier coefficient, respectively.

Since multiplication in the time domain is equivalent to convolution in the frequency domain, the spectrum of $i_{\text{conv},k}$ can be determined as [23]

$$I_{\text{conv},k}^{(m,n)} = (S_{a,k} \otimes I_{a,k})^{(m,n)} + (S_{b,k} \otimes I_{b,k})^{(m,n)} + (S_{c,k} \otimes I_{c,k})^{(m,n)} \quad (7)$$

where $I_{a,k}$, $I_{b,k}$, and $I_{c,k}$ and $S_{a,k}$, $S_{b,k}$, and $S_{c,k}$ are the spectra of the corresponding phase currents and switching functions, respectively. The symbol $\otimes$ denotes 2-D convolution of the Fourier coefficients of the signals, that is

$$(X \otimes Y)^{(m,n)} = \sum_{k_m=-\infty}^{+\infty} \sum_{k_n=-\infty}^{+\infty} X^{(k_m,k_n)} Y^{(m-k_m,n-k_n)}. \quad (8)$$

The spectra of the phase currents are assumed to contain only the reference component, which means that only the components where $m = 0$, $n = \pm 1$ need to be considered in the convolution. Thus, the following equation can be obtained from (7) and (8):

$$\begin{aligned} (S_{a,k} \otimes I_{a,k})^{(m,n)} &= \sum_{k_m=-\infty}^{+\infty} \sum_{k_n=-\infty}^{+\infty} I_{a,k}^{(k_m,k_n)} S_{a,k}^{(m-k_m,n-k_n)} \\ &= I_{a,k}^{(0,-1)} S_{a,k}^{(m,n+1)} + I_{a,k}^{(0,1)} S_{a,k}^{(m,n-1)} \end{aligned} \quad (9)$$

where $I_{a,k}^{(0,-1)} = (\hat{I}/2)e^{j\theta_0}$ and $I_{a,k}^{(0,1)} = (\hat{I}/2)e^{-j\theta_0}$.

Equation (9) provides the spectral content of the contribution of phase leg a to the submodule input current.

The b and c phase contributions are calculated similarly, but with $\theta_0 = 2\pi/3$ and $\theta_0 = -2\pi/3$, respectively. Substituting (9) into (7) provides $I_{\text{conv},k}$. By summing the input currents $i_{\text{conv},k}$ of all submodules, the Fourier coefficients of the summation $\sum_{k=1}^N i_{\text{conv},k}$ can finally be obtained.

Ordinary naturally sampled sine-triangle modulation will now be used as an example. The harmonic coefficients of the switching function generated by this modulation method can be expressed as [24]

$$S_{a,k}^{(0,0)} = 1/2 \quad (10)$$

$$S_{a,k}^{(0,\pm 1)} = M/2 \quad (11)$$

$$S_{a,k}^{(m,n)} = \frac{2}{m\pi} J_n(mM \frac{\pi}{2}) \sin\left((m+n)\frac{\pi}{2}\right), \quad m > 1 \quad (12)$$

where M is the modulation index and J_n is the Bessel function of order n .

Therefore, similar to (6), $\sum_{k=1}^N i_{\text{conv},k}$ can also be expanded as a double Fourier series and the coefficients are obtained as

$$\tilde{A}_{00} = \frac{3MN\hat{I}}{4} \cos \theta_0 \quad (13)$$

$$\tilde{A}_{0n} = \tilde{B}_{0n} = 0, \quad n \geq 1 \quad (14)$$

$$\begin{aligned} \tilde{A}_{mn} &= \frac{\hat{I}}{m\pi} \cos\left((m+n)\frac{\pi}{2}\right) \left[1 + 2 \cos\left(\frac{2n\pi}{3}\right)\right] \\ &\times \left\{ \cos \theta_0 \left[J_{n+1}\left(m\frac{\pi}{2}M\right) - J_{n-1}\left(m\frac{\pi}{2}M\right) \right] \right. \\ &\times \sum_{k=1}^N \cos m\theta_{c,k} + \sin \theta_0 \\ &\times \left. \left[J_{n+1}\left(m\frac{\pi}{2}M\right) + J_{n-1}\left(m\frac{\pi}{2}M\right) \right] \sum_{k=1}^N \sin m\theta_{c,k} \right\} \end{aligned} \quad (15)$$

$$\begin{aligned} \tilde{B}_{mn} &= \frac{\hat{I}}{m\pi} \cos\left((m+n)\frac{\pi}{2}\right) \left[1 + 2 \cos\left(\frac{2n\pi}{3}\right)\right] \\ &\times \left\{ \cos \theta_0 \left[J_{n+1}\left(m\frac{\pi}{2}M\right) - J_{n-1}\left(m\frac{\pi}{2}M\right) \right] \right. \\ &\times \sum_{k=1}^N (-\sin m\theta_{c,k}) + \sin \theta_0 \\ &\times \left. \left[J_{n+1}\left(m\frac{\pi}{2}M\right) + J_{n-1}\left(m\frac{\pi}{2}M\right) \right] \sum_{k=1}^N (\cos m\theta_{c,k}) \right\}. \end{aligned} \quad (16)$$

From (13) and (14), it can be concluded that no fundamental or second-order harmonic is present in the dc link current. Equations (15) and (16) represent the real and imaginary Fourier coefficients of the carrier and sideband harmonics of $\sum_{k=1}^N i_{\text{conv},k}$, respectively. From trigonometry, it follows that the summations $\sum_{k=1}^N (\sin \theta_{c,k})$ and $\sum_{k=1}^N (\cos \theta_{c,k})$ equal zero when $\theta_{c,k} = (2\pi/N)k$. Therefore, in order to maximize the elimination of harmonic components in $\sum_{k=1}^N i_{\text{conv},k}$, the interleaving angle $\theta_{c,k}$ should be selected as $(2\pi/N)k$. Thus, the carrier and corresponding sideband harmonics are eliminated when $m \neq pN$, $p = 1, 2, \dots$

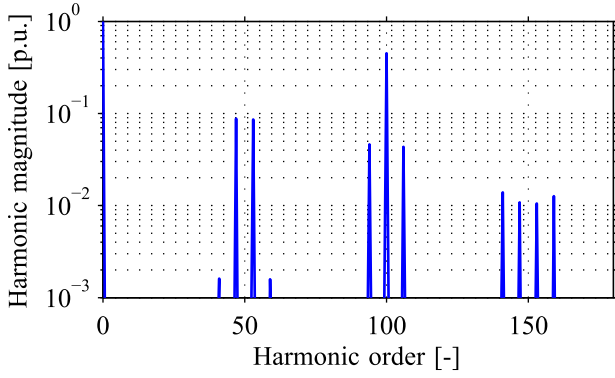


Fig. 2. Calculated dc link current spectrum when the interleaving angle is 0° , $N = 3$, $\cos \theta_0 = 1$, $M = 0.9$, $f_c = 2500$ Hz, and $f_0 = 50$ Hz.

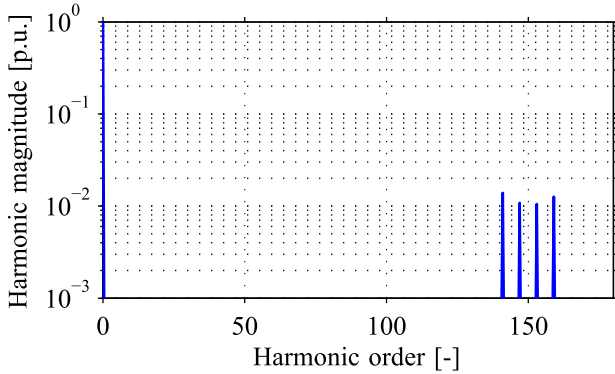


Fig. 3. Calculated dc link current spectrum when the interleaving angle is 120° , $N = 3$, $\cos \theta_0 = 1$, $M = 0.9$, $f_c = 2500$ Hz, and $f_0 = 50$ Hz.

Figs. 2 and 3 compare the impact of the interleaving angle on the resulting dc link current spectrum using (3) (2500 Hz switching frequency is selected for ease of visualization). As seen, the resulting spectra are in agreement with the above discussion.

Due to space limitations, the double Fourier coefficients of more complex PWM schemes, such as space vector PWM (SVPWM), are not presented and analyzed here. In [25], the interleaving impact on the dc link current is investigated for SVPWM-controlled parallel connected three-phase converters and is similar to that of the SPWM-controlled converter.

IV. SWITCHING LOSSES

Power losses and semiconductor junction temperature depend on each other, and thus loss calculation without a proper thermal model is not accurate. In this paper, an electrothermal model, which combines a loss model and a thermal model, is employed to find correct steady-state losses. Using the instantaneous temperature value derived from the thermal model, temperature-dependent parameters, such as ON-state resistance $R_{DS(on)}$, can be calculated. Then, the loss model uses these parameters to compute the power dissipation, which will be delivered to the thermal model to calculate a new temperature for the next iteration. This process continues until the given temperature tolerance is reached.

For power-MOSFET-based converters, the conduction loss calculation is quite straightforward by calculating $I_{drms}^2 R_{DS(on)}$. Therefore, the following analysis will be focused

only on the switching loss calculation. To analyze the switching process of the SPB converter, one phase leg from Fig. 1 is considered. In [26]–[28], analytical loss models for power MOSFETs in buck converters are presented. Similar to a MOSFET-based buck converter model in these references, Fig. 4(a) shows a simplified equivalent circuit of a phase leg, including parasitic inductances in series with the drain and source terminals of the MOSFET and inherent parasitic capacitances C_{gs} , C_{gd} , and C_{ds} , since the parasitic inductances and nonlinear characteristics of the parasitic capacitors significantly affect the switching process in practice. Waveforms for the switching transitions are shown in Fig. 5, where v_{gs} is the gate–source voltage and v_{ds} and i_d are the drain–source voltage and the drain current of the top switch, respectively.

A. Turn-ON Transition

The equivalent circuits at different periods of the turn-ON transition are shown in Fig. 4(b) and (c).

1) *Period I* [t_0 , t_1]: For simplicity, a freewheeling diode is used instead of a MOSFET as the bottom switch during the commutation period. The load current can be treated as a dc current source I_{load} since the commutation time is sufficiently short. Therefore, Fig. 4(a) can be further simplified as Fig. 4(b), in which $L_s = L_{dl} + L_{sl}$ and $L_d = L_{du} + L_{su}$. At t_0 , the constant gate voltage V_{dr} is added to the top switch, charging the input capacitances C_{gd} and C_{gs} . This period ends when the gate voltage has risen to the threshold voltage V_{th} at t_1 . As the top MOSFET is not activated, there is no power loss during this period.

2) *Period II* [t_1 , t_2]: At t_1 , the drain current i_d starts to take over the load current, discharging the drain–source capacitance C_{ds} . Based on the equivalent circuit in Fig. 4(c), the dynamics of the currents and voltages can be obtained on Laplace form

$$v_{ds}(s) = -sL_d i_D(s) - sL_s i_D(s) \quad (17)$$

$$v_{gs}(s) = v_{ds}(s) + v_{gd}(s) \quad (18)$$

$$i_G(s) = (V_{dr} - v_{gs}(s))/R_g = sC_{gd}v_{gd}(s) + sC_{gs}v_{gs}(s) \quad (19)$$

$$i_D(s) = i_d(s) - sC_{ds}v_{ds}(s) - sC_{gd}v_{gd}(s) \quad (20)$$

$$i_d(s) = g_{fs}(v_{gs}(s) - V_{th}) \quad (21)$$

where g_{fs} is the transconductance of a MOSFET and assumed to be constant in this paper. The gate–source voltage v_{gs} is derived using (17)–(21) as

$$v_{gs}(s) = \frac{V_{dr}}{q_1 s^2 + q_2 s + 1} \quad (22)$$

where $q_1 = L_{eq}C_{eq2} + R_g g_{fs} C_{gd} L_{eq}$, $q_2 = R_g C_{eq1}$, $L_{eq} = L_s + L_d$, $C_{eq1} = C_{gd} + C_{gs}$, and $C_{eq2} = C_{gd} - C_{ds}$.

Then the drain current i_d and drain–source voltage v_{ds} can be derived from (21) and (17). Transferring back to time domain gives the solution of $i_d(t)$ and $v_{ds}(t)$. During [t_1 , t_2], the dissipated turn-ON energy is

$$E_{turn_on} = \int_{t_1}^{t_2} v_{ds}(t) i_d(t) dt. \quad (23)$$

This period ends when the drain current reaches the load current I_{load} at t_2 .

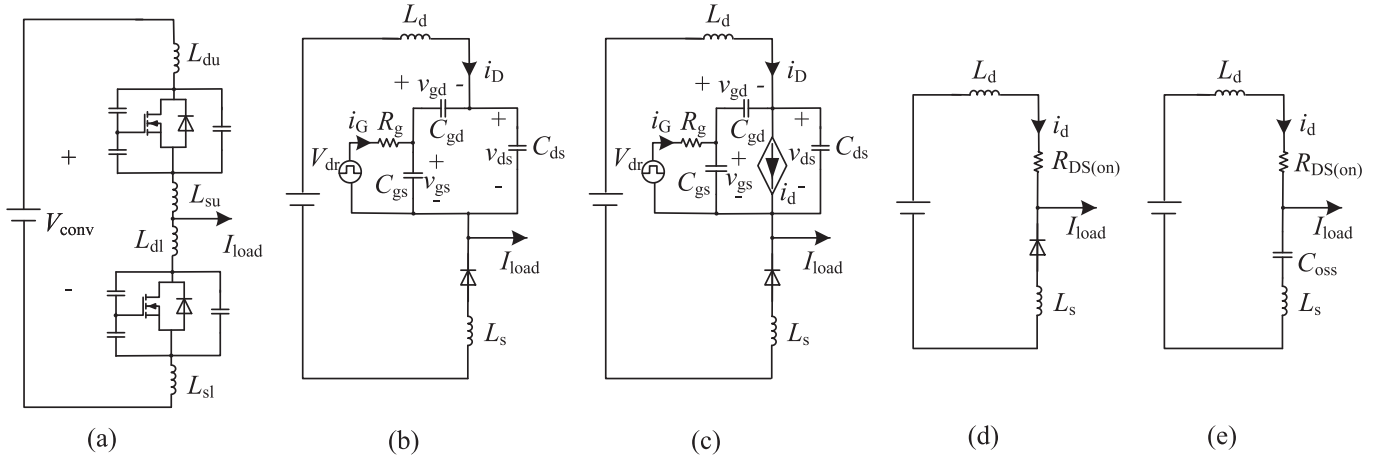


Fig. 4. Simplified equivalent circuits during turn-ON period of the top switch. (a) Equivalent circuit of a phase leg. (b) Period I $[t_0, t_1]$: delay period. (c) Period II $[t_1, t_2]$: turn-ON transition. (d) Period III $[t_2, t_3]$: reverse recovery. (e) Period III $[t_3, t_4]$: ringing.

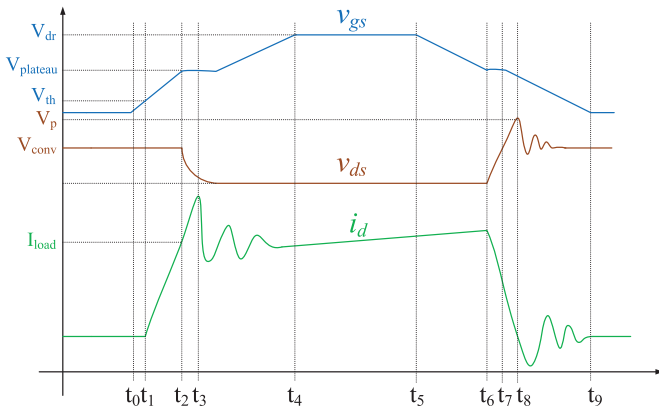


Fig. 5. Key waveforms of top switch during the switching transitions.

3) *Period III* $[t_2, t_4]$: At t_2 , the top switch takes over the load current and the lower diode starts to recover, but still cannot block the voltage. The drain current reaches its peak value at t_3 when the diode starts to block voltage. Beyond t_3 , the ringing continues and we assume it can be completely damped at t_4 . The equivalent circuits for $[t_2, t_3]$ and $[t_3, t_4]$ are shown in Fig. 4(d) and (e), respectively. The drain current during ringing period $[t_3, t_4]$ is expressed using Laplace form

$$i_d(s) = \frac{C_{oss}v_{C_{oss}(0_-)} - sL_{eq}C_{oss}i_{L_{eq}(0_-)}}{L_{eq}C_{oss}s^2 + R_{DS(on)}C_{oss}s + 1} \quad (24)$$

where $v_{C_{oss}(0_-)}$ is the forward voltage of the body diode and $i_{L_{eq}(0_-)} = I_{load}$.

By considering the energy stored at the beginning and the end of $[t_3, t_4]$

$$E_{store_t_3} = \int_{t_2}^{t_3} V_{conv}(i_d(t) - I_{load})dt \quad (25)$$

$$E_{store_t_4} = \frac{1}{2}C_{oss}V_{conv}^2 \quad (26)$$

the dissipated ringing energy is

$$E_{ring_on} = E_{store_t_3} - E_{store_t_4}. \quad (27)$$

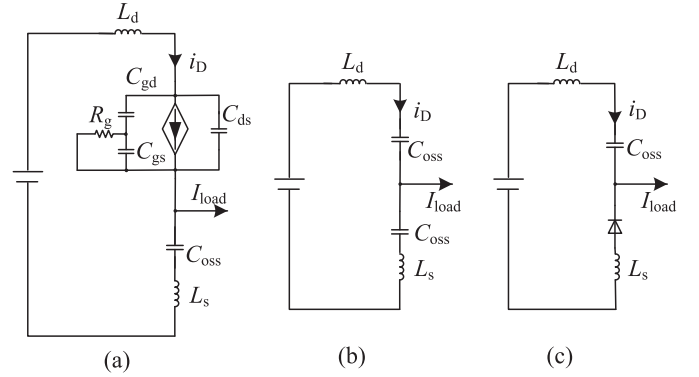


Fig. 6. Simplified equivalent circuits during turn-OFF period of the top switch. (a) Period II $[t_6, t_7]$: turn-OFF transition. (b) Period II $[t_7, t_8]$: v_{ds} continues rising. (c) Period III $[t_8, t_9]$: ringing.

B. Turn-OFF Transition

The equivalent circuits at different periods of the turn-OFF transition are shown in Fig. 6(a)–(c).

1) *Period I* $[t_5, t_6]$: At t_5 , the gate circuit starts to discharge the capacitances C_{gs} and C_{gd} , and the gate–source voltage falls. However, the drain–source voltage and i_D do not change in this period. This period ends at t_6 when v_{gs} decreases to the plateau voltage $V_{th} + I_{load}/g_{fs}$. As this period lasts for a short time and the drain–source voltage v_{ds} does not change, power losses can be neglected during $[t_5, t_6]$.

2) *Period II* $[t_6, t_8]$: At t_6 , the drain–source voltage starts to rise, while the gate voltage is held at the plateau level. The excess current $I_{load} - i_d$ charges the capacitances. i_D does not change until v_{ds} reaches the input voltage V_{conv} at t_7 . After that, v_{ds} continues rising due to the existence of the parasitics. Similar to Section IV-A.2, by solving circuit equations based on Fig. 6(a), the gate–source voltage is

$$v_{gs}(s) = \frac{V_{th} + I_{load}/g_{fs}}{q_1s^2 + q_2s + 1}. \quad (28)$$

This period ends at t_8 when the drain current reaches zero. At this moment, the drain–source voltage reaches its peak

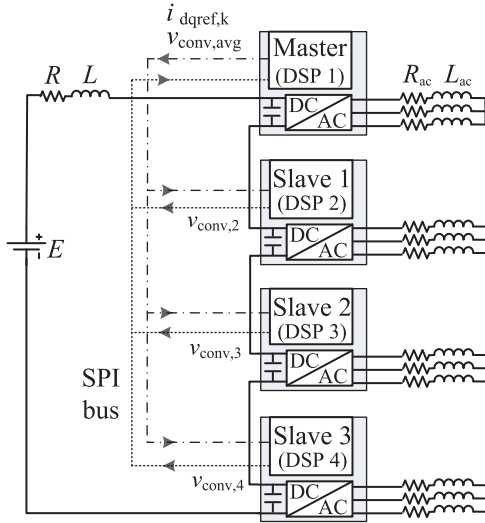


Fig. 7. Schematic of the laboratory setup. The dashed lines mark communication paths.

voltage V_p . During $[t_6, t_8]$, the dissipated turn-OFF energy is

$$E_{\text{turn_off}} = \int_{t_6}^{t_8} v_{ds}(t) i_d(t) dt. \quad (29)$$

3) *Period III* $[t_8, t_9]$: After t_8 , a voltage ringing period continues until v_{ds} reaches its steady-state value at t_9 . Based on the equivalent circuits shown in Fig. 6, the drain current can be derived and is the same as (24) with $v_{C_{oss}(0_-)} = V_{\text{conv}}$ and $i_{L_{eq}(0_-)} = 0$. Similar to Section IV-A3

$$E_{\text{store}_t8} = \frac{1}{2} C_{oss} V_p^2 + \frac{1}{2} L_s I_{\text{load}}^2 \quad (30)$$

$$E_{\text{store}_t9} = \frac{1}{2} C_{oss} V_{\text{conv}}^2 + \frac{1}{2} L_s I_{\text{load}}^2. \quad (31)$$

the losses dissipated in this period is obtained as

$$E_{\text{ring_off}} = E_{\text{store}_t8} - E_{\text{store}_t9} - V_{\text{conv}} \int_{t_8}^{t_9} i_d(t) dt \quad (32)$$

$$= \frac{1}{2} C_{oss} (V_p^2 - V_{\text{conv}}^2) - V_{\text{conv}} \int_{t_8}^{t_9} i_d(t) dt. \quad (33)$$

As mentioned above, the increase in MOSFET on-resistance with junction temperature is taken into account in the thermal model, as is the nonlinearity of the parasitic capacitances with v_{ds} ; these data can be extracted from the datasheet.

V. EXPERIMENTAL VERIFICATION

The designed experiment, from which results are reported in the following, are first used to demonstrate the analytical model used to predict the harmonic content of the dc link current for different interleaving angles. Additionally, the analytical model used to calculate power losses is verified experimentally.

A. Laboratory Prototype

The laboratory prototype and its schematic are shown in Figs. 7 and 8, respectively. It consists of four submodules,



Fig. 8. Experimental setup: four-submodule SPB converter.

TABLE I
PARAMETERS OF THE SETUP UNDER TEST

Parameter	Value	Unit	Symbol
Dc power supply	200	V	E
Number of submodules	4	-	N
Submodule capacitor voltage	50	V	$v_{\text{conv},k}$
Tested ac peak current	6	A	I_0
Submodule dc capacitance	94	μF	C
Switching frequency	20	kHz	f_{sw}
Fundamental frequency	50	Hz	f_0
Measured dc-side resistance	0.091	Ω	R
Measured dc-side inductance	5.5	μH	L
Load resistance	3.2	Ω	R_{ac}
Load inductance	4.67	mH	L_{ac}

each comprising six MOSFETs and a small dc link capacitor. The prototype is connected to four sets of three-phase passive loads. Each submodule is rated at dc 75 V and ac 20 A with a 94 μF dc capacitor and equipped with a control card based on the TMS320F28069 DSP. One of the submodules also contains the master control. The serial peripheral interface protocol is utilized for the communication between the submodules. Each submodule is identical so that in the case of a fault within the master submodule, a slave submodule can replace the faulty master. The dc capacitor voltage measurement is taken and two-phase current sensors are mounted on each submodule.

B. Experimental Results—Harmonic Analysis

To test the validity of the theory in Section III, experimental tests were done using SPWM and three of the four submodules (due to the availability of only three identical loads). The setup parameters are reported in Table I. The experimental dc link current spectra were obtained using a fast Fourier transform analysis of the measured waveforms. Additionally, a MATLAB/Simulink¹ model of the experimental setup was implemented to provide an additional comparison with the analytical prediction and corresponding experimental results.

¹MATLAB and Simulink are registered trademarks of The Mathworks Inc., Natick, MA.

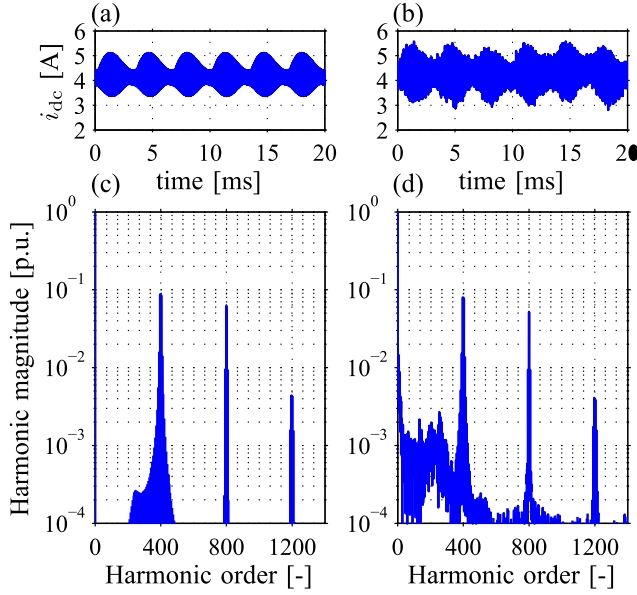


Fig. 9. Simulated and measured dc link currents and their spectra when the interleaving angle is 0° . (a) Simulated i_{dc} . (b) Measured i_{dc} . (c) Spectrum of the simulated i_{dc} . (d) Spectrum of the measured i_{dc} .

1) *Spectrum of the DC Link Current With 0° Interleaving Angle:* Fig. 9 shows the simulated and measured dc link currents i_{dc} and their spectra when the interleaving angle is 0° . The y-axis represents the amplitudes of the harmonic components normalized based on the dc component and the x-axis shows harmonic orders. Note that in Fig. 9(d), the low-order harmonics were caused by the measurement and the harmonics around 12.5 kHz (order 250) were caused by an LC oscillation in the dc side. The pulse number is 400 (f_{sw}/f_0) in this case, and therefore, the carrier harmonics and the corresponding sidebands of i_{dc} appear around harmonic orders 400, 800, 1200, and so on, i.e., $m = 1, 2, \dots$. Considering the factor $\cos((m+n)(\pi/2))$ in (15) and (16), $\tilde{A}_{mn}$ and $\tilde{B}_{mn}$ are nonzero only when $m+n$ is even. Considering the factor $1+2\cos(2n\pi/3)$, $\tilde{A}_{mn}$ and $\tilde{B}_{mn}$ are nonzero only when $n = \pm 3p$, $p = 1, 2, \dots$, and so on. For instance, for $m = 1$, i.e., harmonic order 400, the carrier harmonic should be zero as $m+n = 1$. The corresponding sidebands should only be nonzero when $n = \pm 3, 9, 15$, and so on. For better visualization and further verification, harmonic components around the order 400, 800, and 1200 ($m = 1, 2$, and 3) are depicted in Fig. 10, comparing the theory, simulation, and experiment. There is good agreement between the three results, providing a high degree of confidence in the analytical conclusion. It should be commented that the measurement error of dc side passive parameters may have caused some of the differences between the experiment and the other two.

2) *Spectrum of the DC Link Current With 120° Interleaving Angle:* Fig. 11 shows the simulated and measured dc link currents i_{dc} and their spectra when the interleaving angle is 120° . Obviously, the current ripple in this case is significantly reduced compared with Fig. 9, which is to be expected from the analysis. Fig. 12 further validates that the carrier harmonics and corresponding sidebands of i_{dc} appear only at $m = pN$ for $p = 1, 2, \dots$

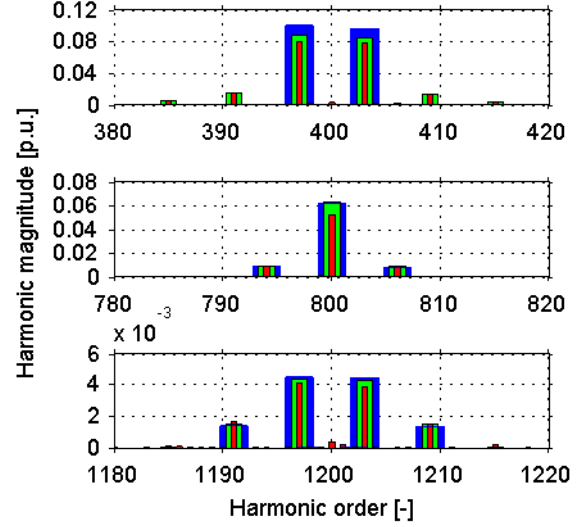


Fig. 10. Harmonic components around 20, 40, and 60 kHz when the interleaving angle is 0° . Blue curve: theory. Green curve: simulation. Red curve: experiment.

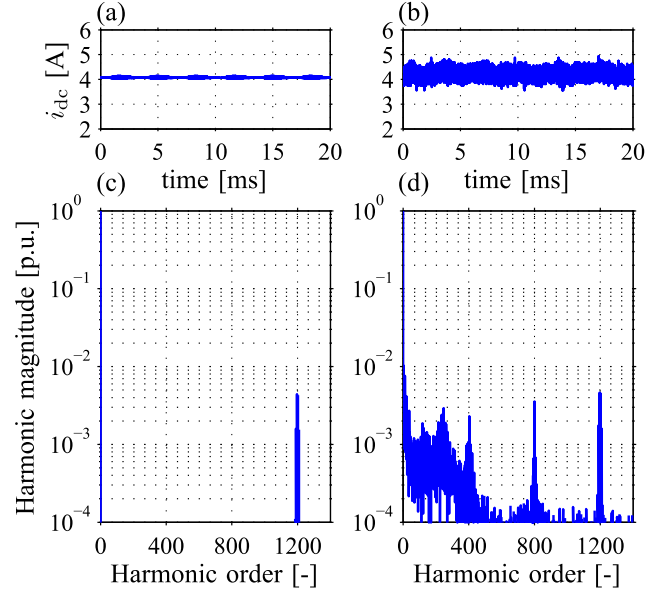


Fig. 11. Simulated and measured dc link currents and their spectra when the interleaving angle is 120° . (a) Simulated i_{dc} . (b) Measured i_{dc} . (c) Spectrum of the simulated i_{dc} . (d) Spectrum of the measured i_{dc} .

C. Experimental Results—Power Losses

To experimentally test the loss model, laboratory measurements were carried out on one submodule of an SPB converter prototype. In this converter, the selected MOSFET is Infineon IPP075N15N3G (150 V/100 A). The submodule was tested at a fixed load of 2Ω , 4.67 mH Y-connected per phase, with different switching frequencies and output currents.

As shown in Fig. 13, the analytical model provides accurate loss results and that the loss trends with regard to output current and switching frequency are clearly reproduced by the model.

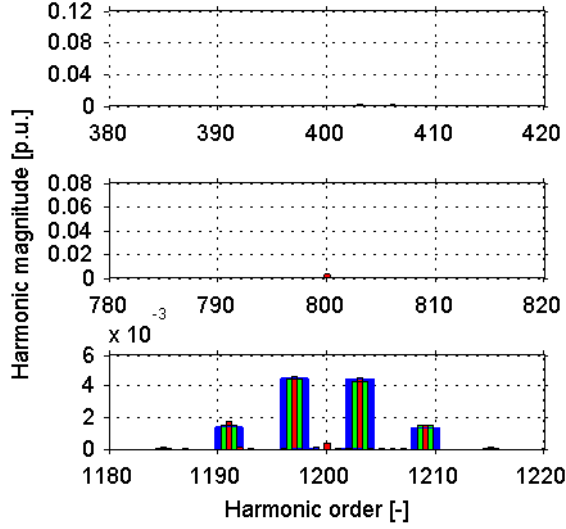


Fig. 12. Harmonic components around 20, 40, and 60 kHz when the interleaving angle is 120°. Blue curve: theory. Green curve: simulation. Red curve: experiment.

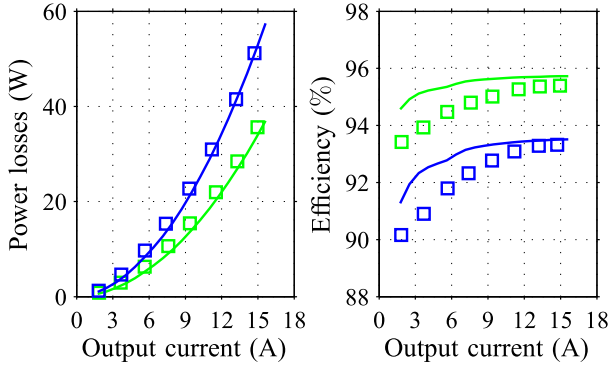


Fig. 13. Comparison of the calculated power losses and measured power losses. Green curve: a 20-kHz switching frequency was used. Blue curve: a 40-kHz switching frequency was used. Solid line: calculated power losses and efficiency. Squares: measured power losses and efficiency.

VI. AUTOMOTIVE APPLICATION

The SPB converter can be a potential candidate for automotive applications. Based on the analytical model in Sections III and IV, the SPB converter is compared with a conventional two-level three-phase converter based on a typical automotive drive system. The main parameters are listed in Table II [29].

According to (3), the dc link peak-to-peak current ripple as a function of switching frequency and dc link capacitance can be calculated. In Fig. 14, the dc link peak-to-peak current ripple ΔI_{ripple} and the energy storage capability τ are defined as

$$\Delta I_{\text{ripple}} = \frac{I_{\text{dc}}(\omega)|_{\text{ac}}}{I_{\text{dc}}|_0} \times 100 \quad (34)$$

$$\tau = \frac{CV_{\text{dc}}^2}{2NP} \quad (35)$$

where $I_{\text{dc}}|_0$ represents the dc component in i_{dc} , $I_{\text{dc}}|_{\text{ac}}$ represents harmonic components in i_{dc} , and P is the total active power. For conventional two-level converter, $N = 1$.

TABLE II
MAIN PARAMETERS OF THE DRIVE SYSTEM FOR BENCHMARKING

Parameter	Value	Unit	Symbol
Dc power supply	400	V	E
Number of submodules	4	-	N
Submodule capacitor voltage	100	V	$v_{\text{conv},k}$
Rated phase peak current	137	A	I_0
Rated speed	4000	rpm	n
Phase resistance	0.0165	Ω	R
d-axis inductance	0.41	mH	L_d
q-axis inductance	0.75	mH	L_q
Rated torque	85	Nm	T_{em}
Power factor	0.794	-	$\cos \theta_0$

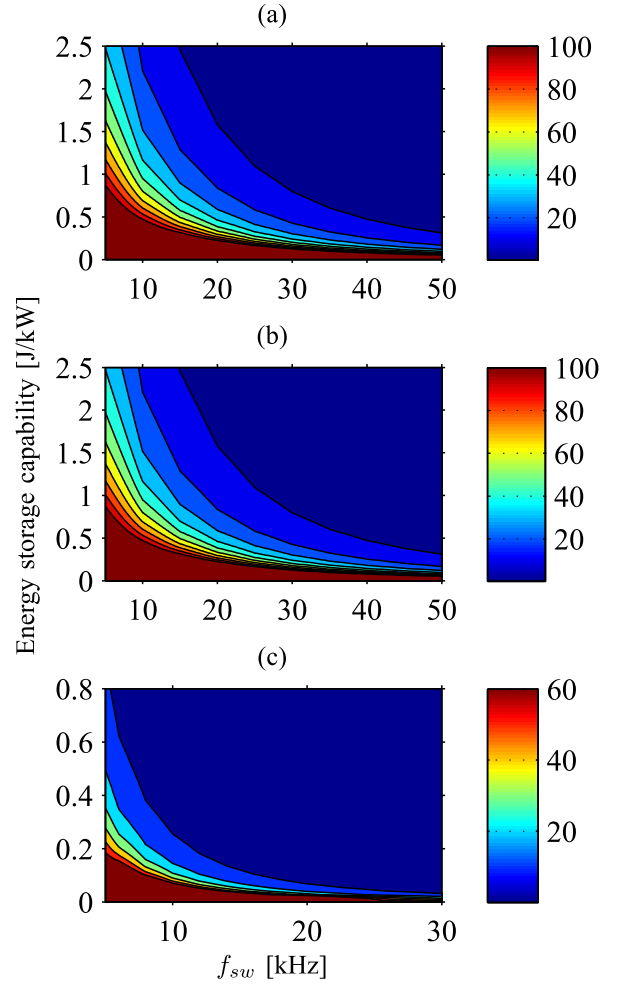


Fig. 14. DC link peak-to-peak current ripples (in percentage, normalized with respect to the dc component of i_{dc}) as a function of switching frequency and energy storage capability at rated power ($L_{\text{bat}} = 1 \mu\text{H}$). (a) Two-level three-phase converter. (b) SPB converter without interleaving. (c) SPB converter with interleaving.

The electrical model of a Li-ion battery reported in [30] is used with a total battery resistance of 100 m Ω and a total battery inductance of 1 μH in the analytical model. Comparing Fig. 14(a) and (b), the current ripple maps from the two-level converter and the SPB converter are identical because the SPB concept has strong similarities with the

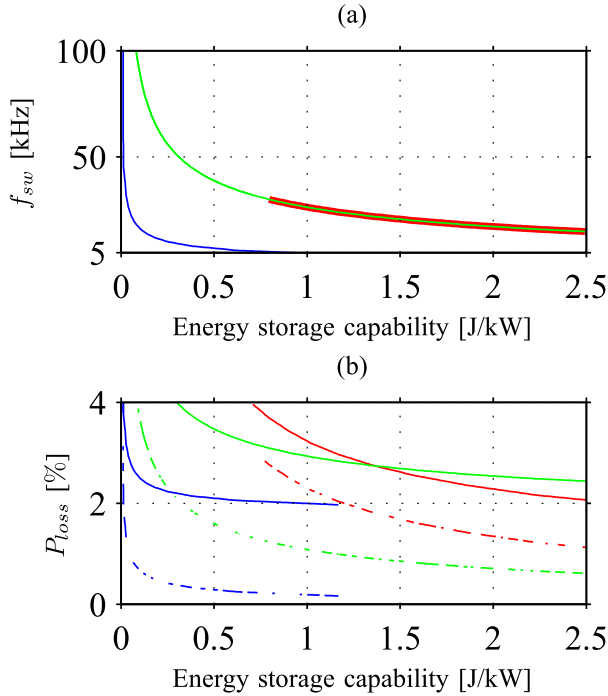


Fig. 15. Comparison of power losses of the SPB converter and the two-level three-phase converter as a function of energy storage capability at rated power with $\Delta I_{ripple} = 10\%$. Red curve: two-level three-phase converter. Green curve: the SPB converter without interleaving. Blue curve: the SPB converter with interleaving. Solid line: total losses. Dashed-dotted line: switching losses.

conventional two-level converter. Indeed, the topology degenerates into a two-level converter when the number of submodules equals one. The current ripple can be reduced by increasing either the energy storage capability of the dc link capacitor or switching frequency. Moreover, by adopting the interleaving modulation, the current ripple in the SPB converter can be further reduced by up to 80%.

For conventional two-level converters, the switching frequency of the semiconductor devices (silicon IGBT) required to handle the ac current and dc voltage is typically limited to below 30 kHz. The selected IGBT module for comparison is Infineon FS200R07PE4 (650 V/200 A) and the corresponding power losses are calculated based on the method in [31]. As seen in Fig. 15(a), this implies that a fairly large and bulky dc link capacitor is required. For instance, a smoothing capacitor with an 880 μ F (1.17 J/kW) capacitance is used in the 60-kW drive system of the 2010 Toyota Prius car [32]. The volume requirement of the dc link capacitor in the SPB converter is rather small. A 100 μ F (0.27 J/kW) capacitor along with the 50 kHz switching frequency is enough to keep the dc link peak-to-peak current ripple ΔI_{ripple} less than 10%.

However, high switching frequency results in high power losses. As seen in Fig. 15(b), more than 5% power will be dissipated in the SPB converter if no optimized modulation technique applies in the high switching frequency region (>50 kHz). On the other hand, with interleaving modulation, 40% lower switching frequency is enough to keep the same current ripple and to use the same size of the dc link capacitor.

Consequently, the power losses can be significantly reduced in the SPB converter with fairly small dc capacitance.

VII. CONCLUSION

In this paper, analytical models to predict the dc link current harmonic content and switching losses of the SPB converter were presented and experimentally verified. Based on these theoretical analyses, the SPB converter was compared with a two-level three-phase converter in a typical 35 kW automotive application, considering the dc link capacitor, current ripple, as well as power losses.

In the high switching-frequency region, the SPB converter exhibits superiority in terms of current ripple and the size of the dc link capacitor. This makes compact integration of an SPB converter and a corresponding electric machine easier. On the other hand, the effects of the modulation scheme on the resulting dc link current predicted that some carrier harmonics and corresponding sidebands of the dc-link current can be eliminated by properly phase shifting (interleaving) the initial angles of the carriers among the submodules. Besides the benefit from the converter allowing for the use of low-voltage high switching frequency semiconductors, interleaving can reduce the dc link current ripple up to 80%. Compared with the noninterleaved SPB converter, the switching frequency can be decreased with the same size of the dc link capacitor, thus reducing the power losses. It should also be noted that the interleaving generally adds complexity to the main circuit and control system, and therefore, a tradeoff between the modulation, power losses, and the size of passive components should be taken into account in the design process.

An important issue for future work is the integration of the SPB converter and the electric machine. Apart from the electromagnetic analysis, this paper should also take cost and thermal analysis into account.

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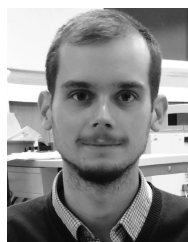
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